Fast Export Manual

Written By Erich MacLean

03-2023, SCANeR Studio 1.6

Intended for use in the DRiVE Lab

Preface:

The Data Pipeline project was created in order to better optimize data collection from the SCANeR Studio 1.6 simulation software. The manual details the use and implementation of the Fast Export method, designed to extract data directly into a .csv file while simulations are active.

Export Configurator:

In order to collect data fields from SCANeR Studio, a program exists to configure which fields of data are collected and stored in the .csv file. This program can be found on the desktop labeled *export.exe* and has the following symbol:

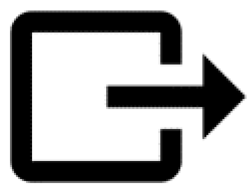


Figure 1: export.exe

The program will open to a screen that looks similar to the following:

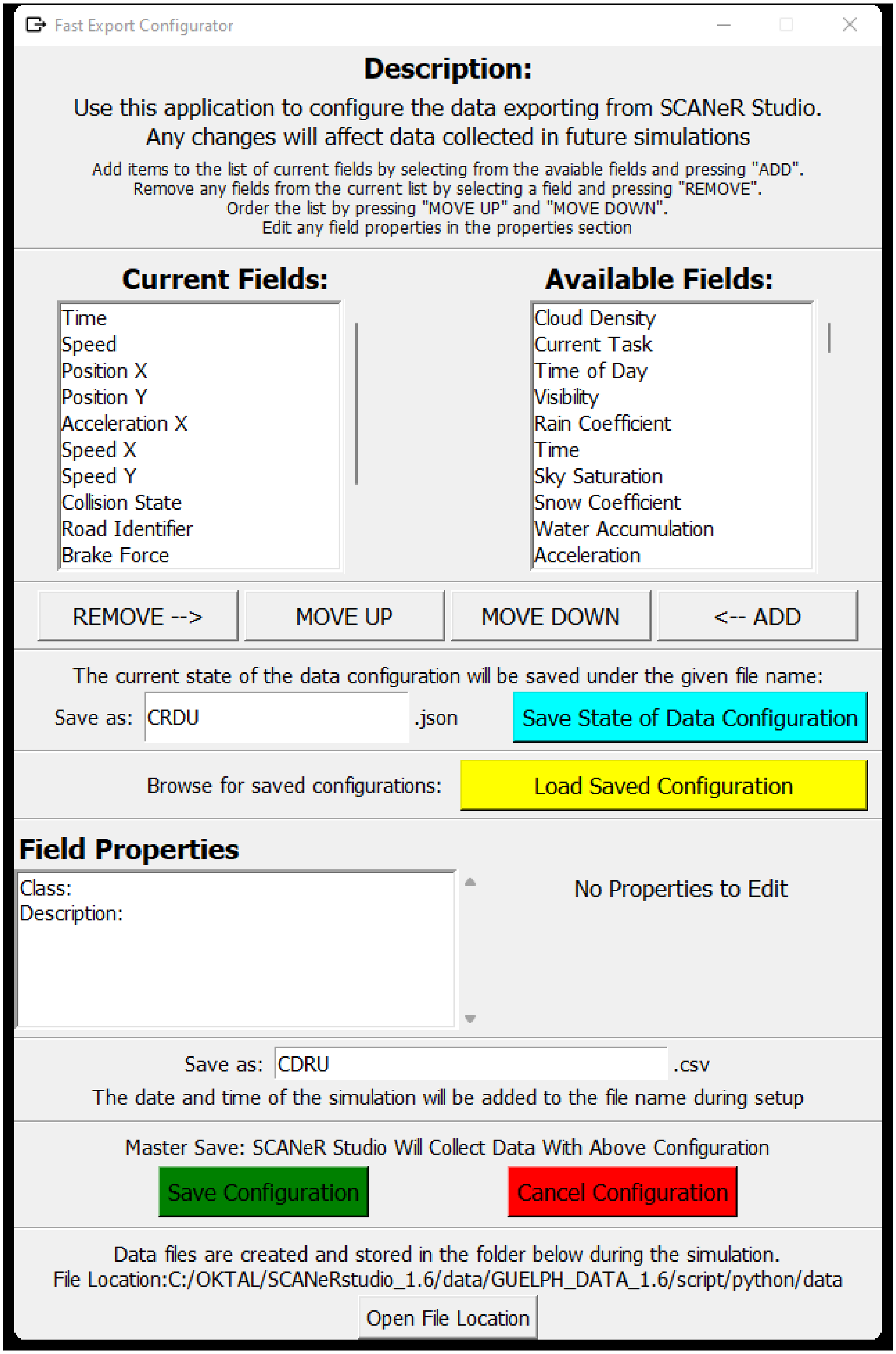


Figure 2: Sample view of the Fast Export Configurator

Description:

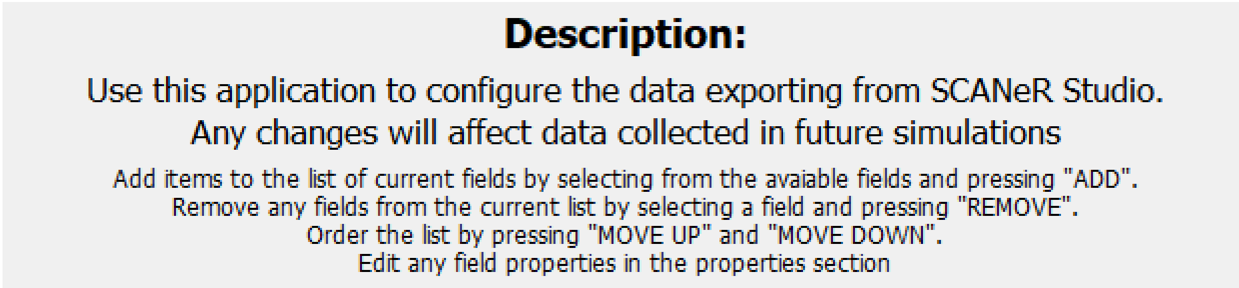


Figure 3: Description section of the window

At the top of the window, there is a description of the tool, along with basic instructions on how to operate it.

Fields:

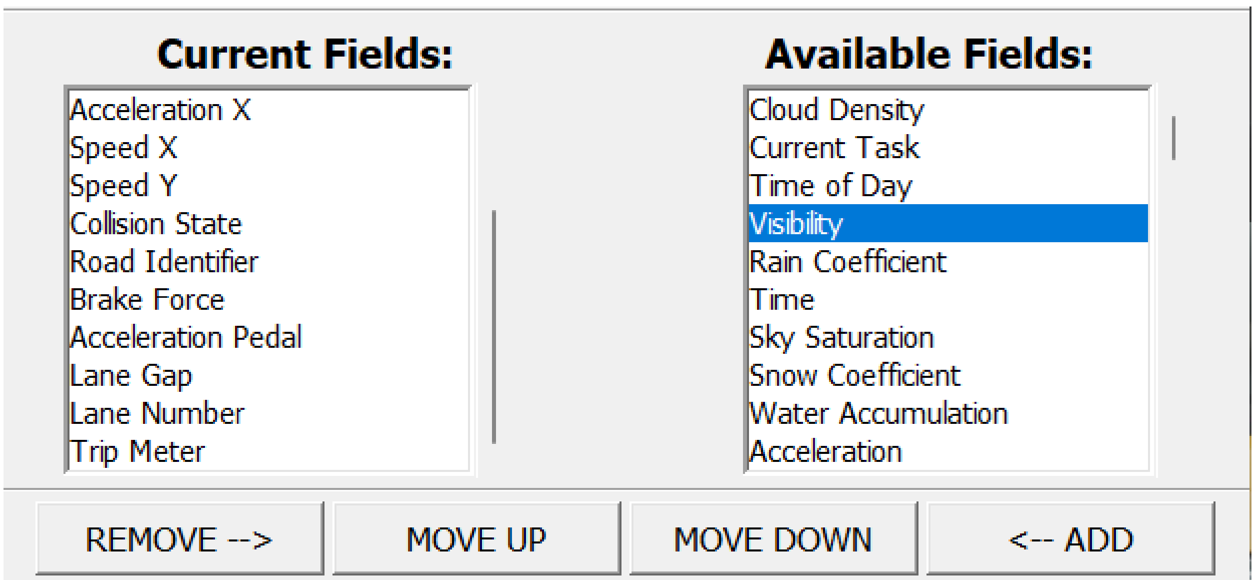


Figure 4: List of current fields is shown on the left, and a list of available fields is shown on the right. Operating buttons are below these boxes

The field lists are where the data that is to be collected is selected. The current fields on the left are the list of data that will be collected during a driving simulation. The available fields on the right is the selection list that contains everything currently functional in the exporting tool.

To add a field to the current fields, select a field from the right, and click "🡐ADD". To remove a field, select a field from the left, and click "REMOVE →". To reorder the fields, select a field on the left, and click "MOVE UP" or "MOVE DOWN". The field at the top of the list will be in the first column in the csv file.

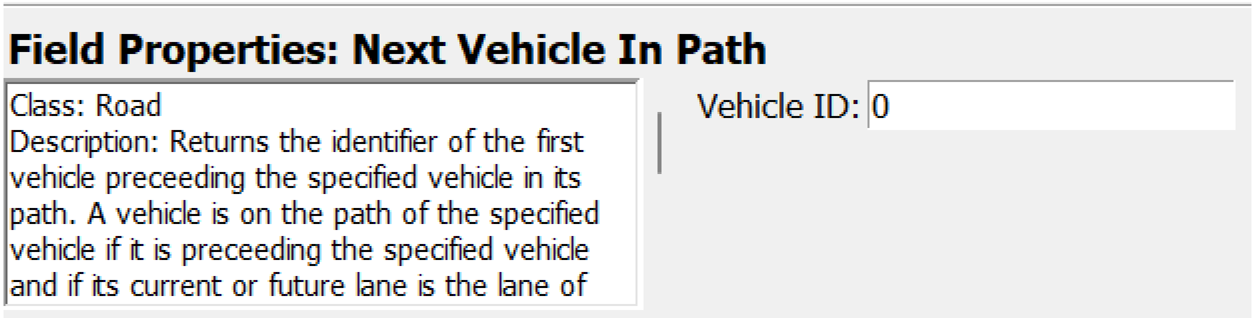


Figure 5: Field description and properties

The field description and properties can be viewed and edited after selecting a field from either list. Some do not have a field and will indicate as such.

Saving a List of Fields:

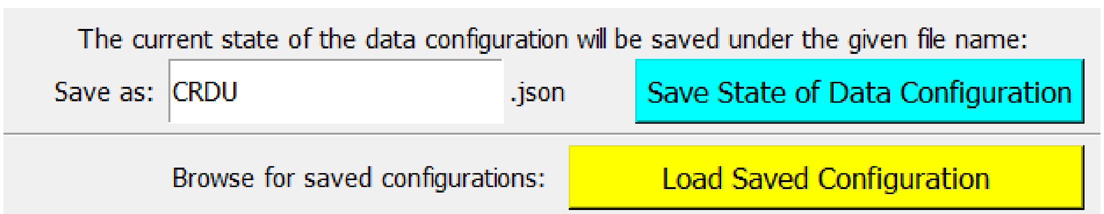


Figure 6: Configuration Saving

In an update to the previous version, functionality ahs been added to allow for the saving of multiple field configuration files. The fields that are saved with this are located in the list of Current Fields. The name of the file is entered on the left. When saving, ensure that this name is changed.

To save the list of Current Fields, click the Save State of Data Configuration button. This will save it to a file and prompt if the file already exists. Do not overwrite files that are not yours. Please check the name is different prior to saving.

To load a configuration, select Load Saved Configuration. This will populate the Current Fields list, and override anything in that list currently. **When running a study, ensure that the correct list is loaded in the Current Fields section BEFORE running a simulation.**

Also note that this is NOT the final step in setting up for data exporting. This feature is so that different configurations can be used.

Preparing for Exporting:

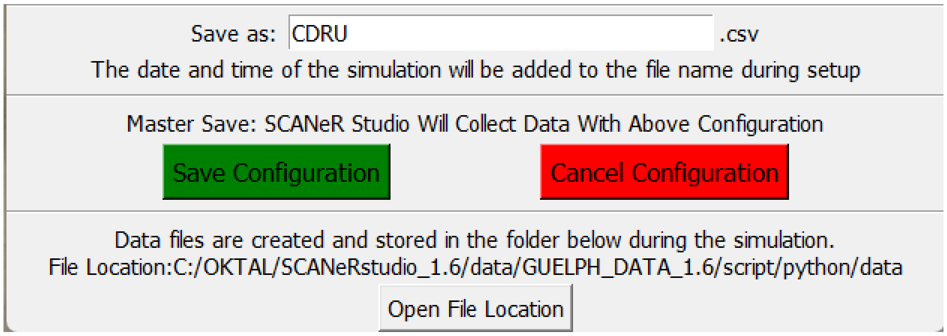


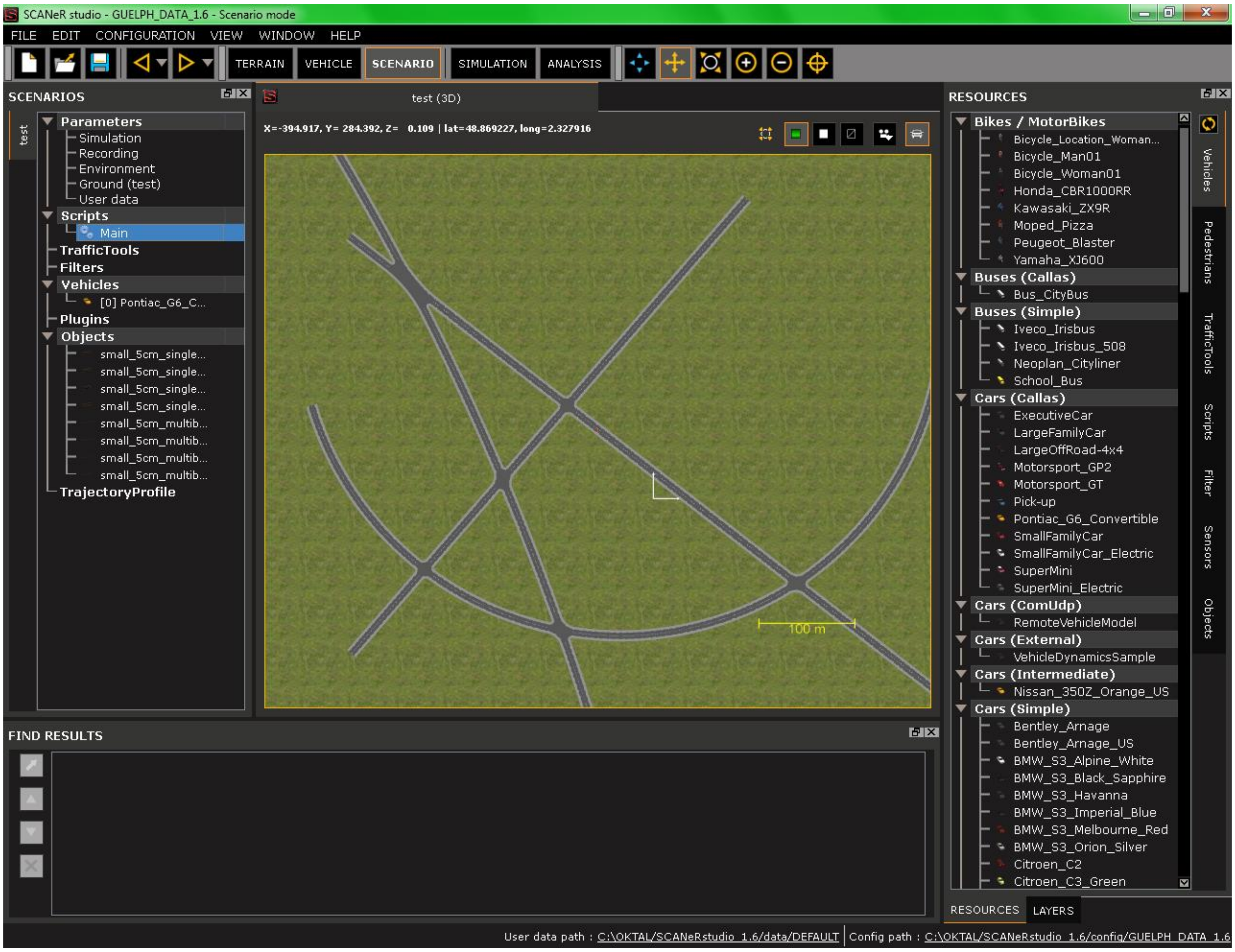
Figure 7: CSV Naming and Saving

BEFORE running a simulation, please ensure that the correct configuration is loaded for data collection. The data will be stored in a CSV file with the name specified in the bottom half of the window. Please indicate the name of the file that you would like in this section. The date and time are appended to the file name at the time of the simulation.

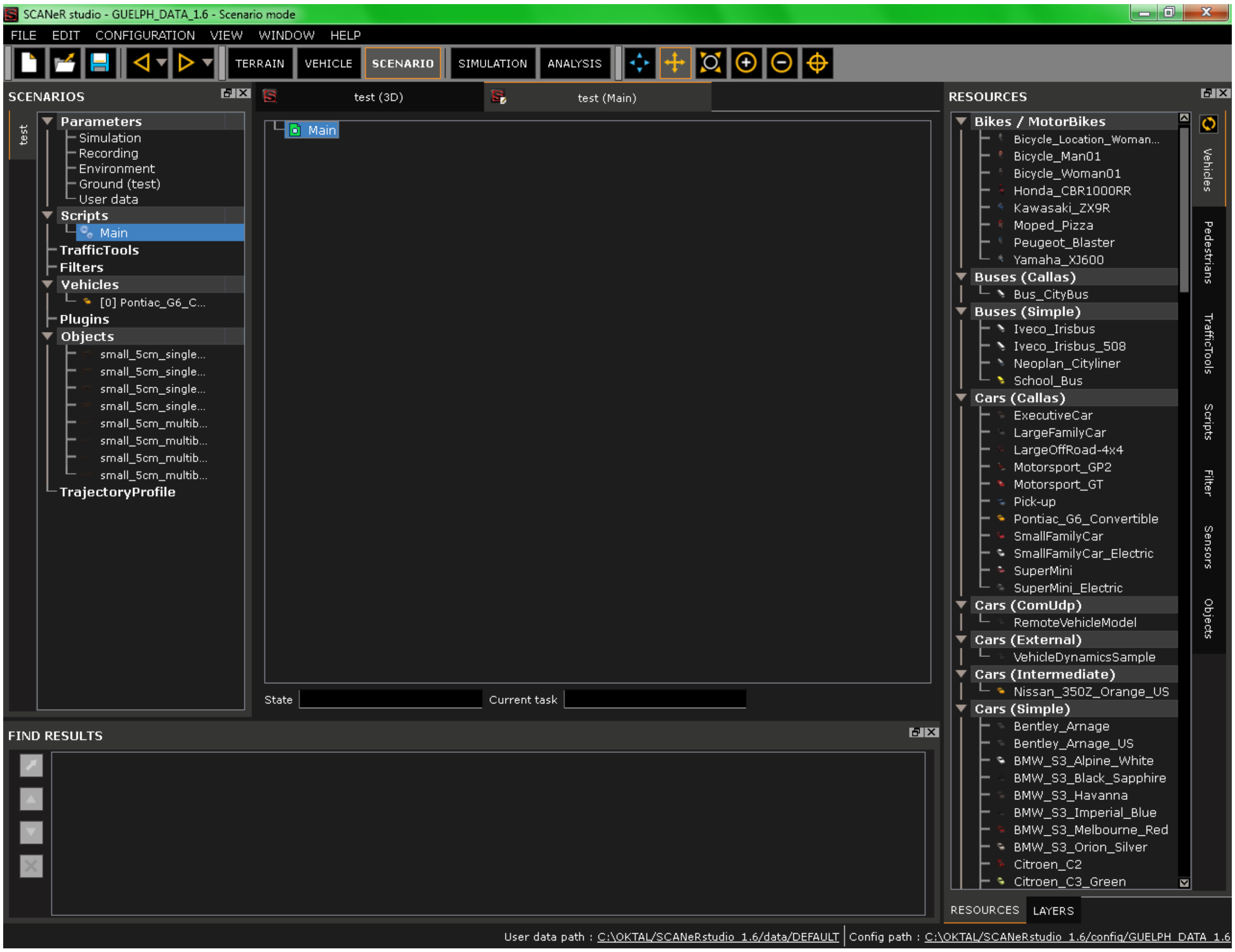
Pressing Save Configuration will save the Current Fields configuration for the fields that will be exported from SCANeR. It will also register the name of the CSV file that the data will be stored in.

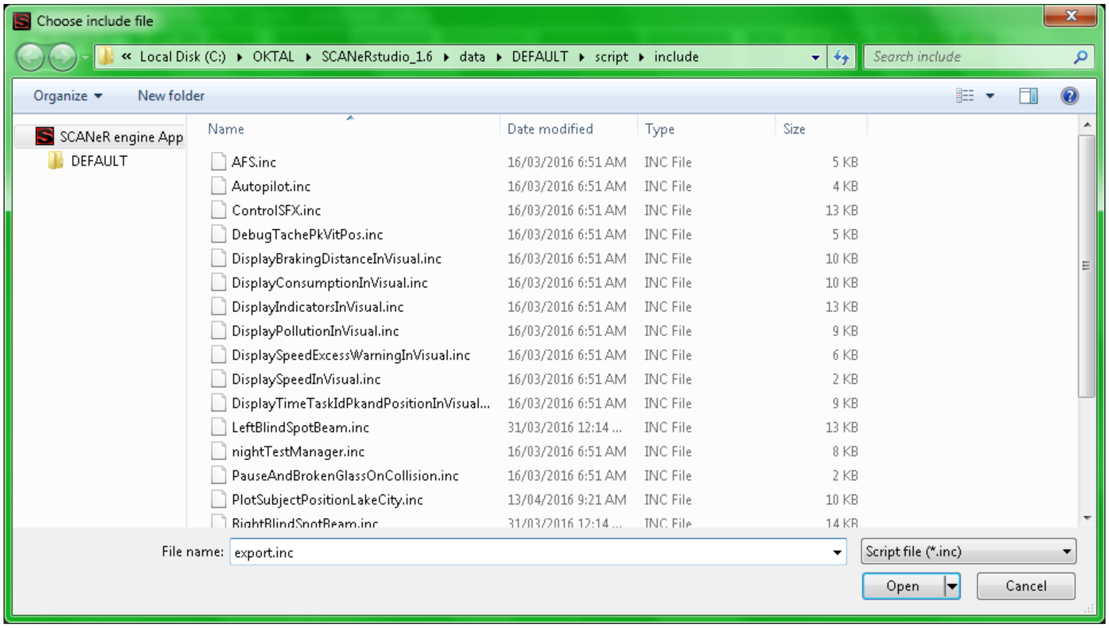
SCANeR Studio:

Once the desired configurations are created, there is a simple process to set up a scenario script to export the data. This is a sample scenario window. Click on the Main script in the left side of the window to open the script for the scenario.

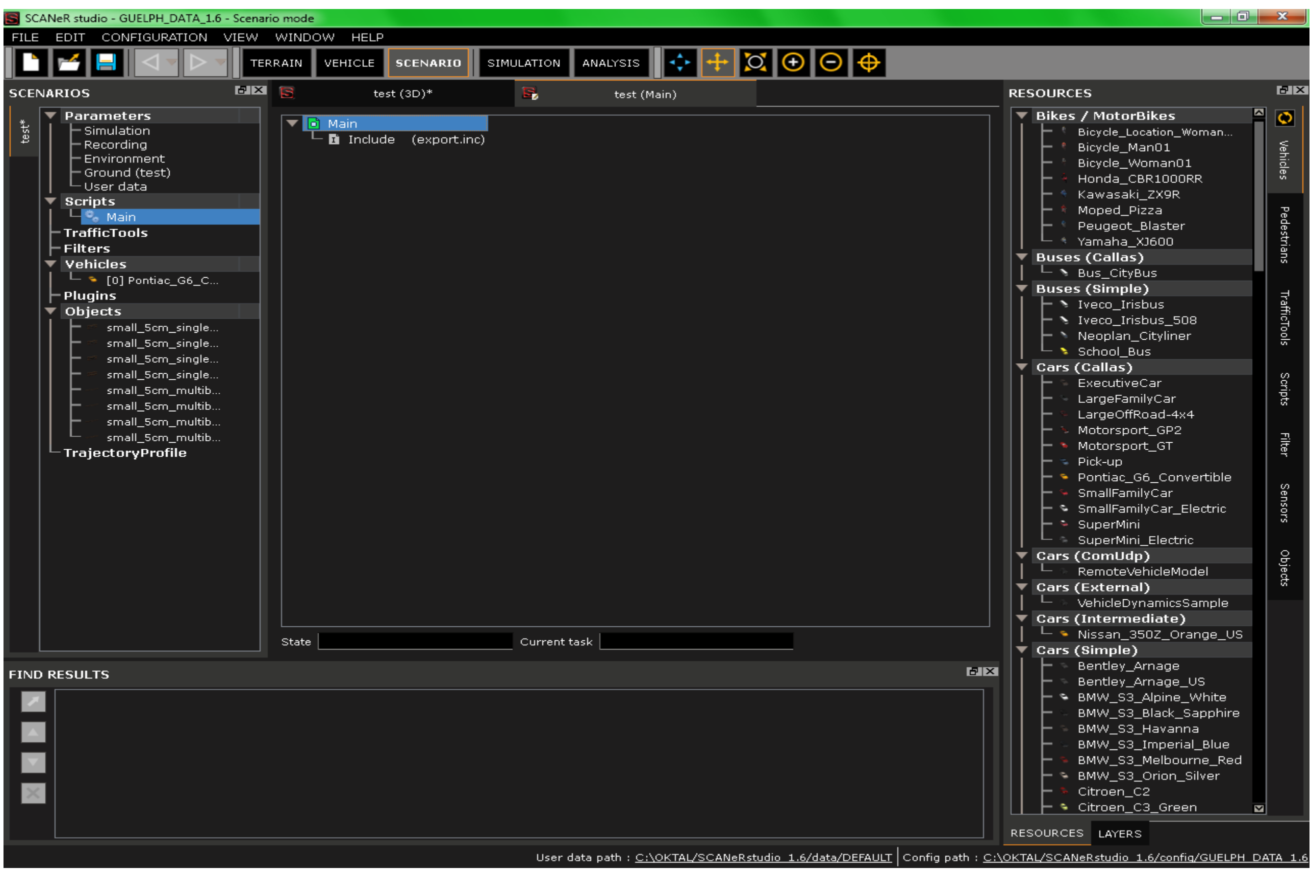


Once the script is open, navigate to the top of the script and right click *Main*. Then select the option to *Add Include*. The second image below shows the window that opens. Chose the file named *export.inc*.





Once the file is selected, the include will appear at the top of the script. any user code can be added underneath without any impact.



Notes:

A. **The “Channel” field is currently non-operational.** Due to the way SCANeR Studio works, the exportChannel can only be called within the script itself and not from a python script execution. This means that in order to export custom variables, one has to save the data into a variable and then get that variable. This will be implemented in upcoming updates.

B. **There is a json file that contains a list of all the default field collectors.** This directory is one level up from the data directory and is named defaults.json. For each field, there is a name, SCANeR function using the mice command, class, description, argument count and argument variables. Do not edit unless you know what you are doing.

C. **There is now a folder that contains all the different configuration files for all users.** Do not delete this folder.

D. **Be careful of Tasks.** The functionality of an include alongside scenario Tasks has not been fully tested.

E. **As of 08-2022, Erich MacLean is the current lab technician managing this product.** Should there be any issues using this tool, or updates requested, please contact emacle05@uoguelph.ca for information and assistance.